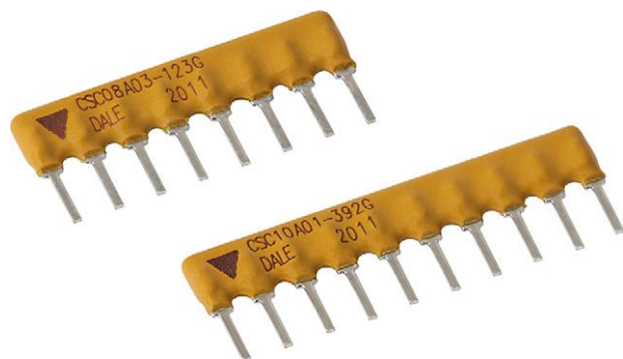


Thick Film Resistor Networks, Single-In-Line, Conformal Coated SIP



FEATURES

- Isolated, bussed, and dual terminator schematics available
- Body height: "A" profile = 0.195" (4.95 mm) and "B" profile = 0.295" (7.50 mm) standard; custom "C" profile = 0.350" (8.89 mm) also available
- "A" profile standard in 4 thru 12 pins
- Thick film resistive elements
- Reduces total assembly costs
- Resistor elements protected by tough epoxy conformal coating
- Wide resistance range (10 Ω to 2.2 M Ω)
- Available in bulk pack as standard; optional tube pack is also available
- Meets EIA/ECA-CB23 rev. G whisker test requirements for class 1A products
- Material categorization: for definitions of compliance please see www.vishay.com/doc?99912



Available


RoHS*
Available

Note

* This datasheet provides information about parts that are RoHS-compliant and / or parts that are non RoHS-compliant. For example, parts with lead (Pb) terminations are not RoHS-compliant. Please see the information / tables in this datasheet for details

STANDARD ELECTRICAL SPECIFICATIONS							
GLOBAL MODEL / SCHEMATIC	PACKAGE HEIGHT	POWER RATING ELEMENT ⁽¹⁾ $P_{70^{\circ}\text{C}}$ W	RESISTANCE RANGE Ω	TEMPERATURE COEFFICIENT (-55 $^{\circ}\text{C}$ to +125 $^{\circ}\text{C}$) $\pm$ ppm/ $^{\circ}\text{C}$	TOLERANCE ⁽²⁾ $\pm$ %	TCR TRACKING ⁽¹⁾ (-55 $^{\circ}\text{C}$ to +125 $^{\circ}\text{C}$) $\pm$ ppm/ $^{\circ}\text{C}$	MAX. WORKING VOLTAGE ⁽³⁾ V_{DC}
CSCxxx01	A	0.20	10 to 50	250	1, 2, 5	50	100
			50.1 to 2.2M	100			
	B	0.25	10 to 50	250			
			50.1 to 2.2M	100			
CSCxxx03	A	0.30	10 to 50	250	1, 2, 5	50	100
			50.1 to 2.2M	100			
	B	0.40	10 to 50	250			
			50.1 to 2.2M	100			
CSCxxx05	A	0.20	10 to 50	250	1, 2, 5	150	100
			50.1 to 2.2M	100			
	B	0.25	10 to 50	250			
			50.1 to 2.2M	100			

Notes

- See derating curves for package power rating
- ⁽¹⁾ For resistor power ratings at +25 $^{\circ}\text{C}$ see derating curves
- ⁽²⁾ $\pm$ 2 % standard, $\pm$ 1 % and $\pm$ 5 % available
- ⁽³⁾ Continuous working voltage shall be $\sqrt{P \times R}$ or maximum working voltage, whichever is less

GLOBAL PART NUMBER INFORMATION

New Global Part Numbering: **CSC08A03100RGEK** (preferred part number format)

C	S	C	0	8	A	0	3	1	0	0	R	G	E	K			
GLOBAL MODEL	PIN COUNT		PACKAGE HEIGHT		SCHEMATIC		RESISTANCE VALUE		TOLERANCE CODE		PACKAGING		SPECIAL				
CSC	04 to 12 pin available 04 = 4 pin 08 = 8 pin 12 = 12 pin		A = "A" profile B = "B" profile		01 = bussed 03 = isolated 00 = special		R = Ω K = kΩ M = MΩ 10R0 = 10 Ω 680K = 680 kΩ 1M00 = 1.0 MΩ 0000 = 0 Ω Jumper		F = ± 1 % G = ± 2 % J = ± 5 % S = special Z = 0 Ω Jumper		EK = lead (Pb)-free, bulk PA = tin / lead, bulk		Blank = standard (dash number) (up to 3 digits) From 1 to 999 as applicable				

Historical Part Number example: **CSC08A03101GEK** (will continue to be accepted)

CSC	08	A	03	101	G	EK
HISTORICAL MODEL	PIN COUNT	PACKAGE HEIGHT	SCHEMATIC	RESISTANCE VALUE	TOLERANCE CODE	PACKAGING

New Global Part Numbering: **CSC08A05131AGEK** (preferred part number format)

C	S	C	0	8	A	0	5	1	3	1	A	G	E	K			
GLOBAL MODEL	PIN COUNT		PACKAGE HEIGHT		SCHEMATIC		RESISTANCE VALUE		TOLERANCE CODE		PACKAGING		SPECIAL				
CSC	04 to 12 pin available 04 = 4 pin 08 = 8 pin 12 = 12 pin		A = "A" profile B = "B" profile		05 = dual terminator		3 digit impedance code, followed by alpha modifier (see impedance table)		F = ± 1 % G = ± 2 % J = ± 5 %		EK = lead (Pb)-free, bulk PA = tin / lead, bulk		Blank = standard (dash number) (up to 3 digits) From 1 to 999 as applicable				

Historical Part Number example: **CSC08A05131AGEK** (will continue to be accepted)

CSC	08	A	05	221	331	G	EK
HISTORICAL MODEL	PIN COUNT	PACKAGE HEIGHT	SCHEMATIC	RESISTANCE VALUE 1	RESISTANCE VALUE 2	TOLERANCE CODE	PACKAGING

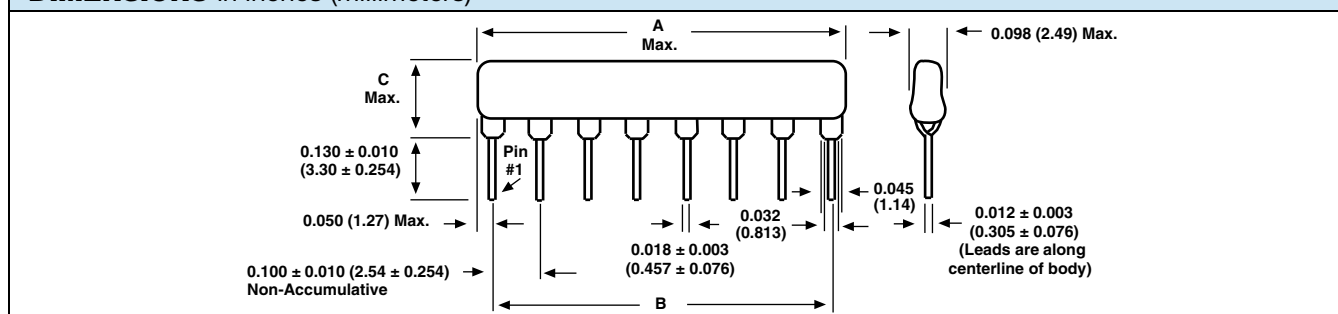
Note

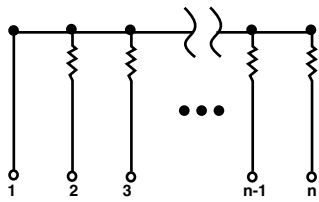
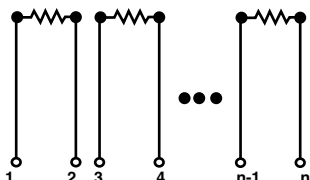
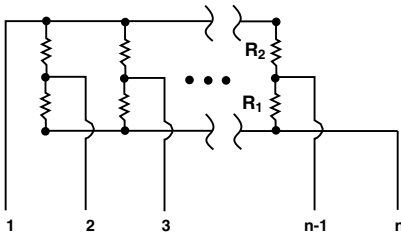
- For additional information on packaging, refer to the Through-Hole Network Packaging document (www.vishay.com/doc?31542)

TECHNICAL SPECIFICATIONS

PARAMETER	UNIT	CSC SERIES
Voltage coefficient of resistance	V_{eff}	< 50 ppm typical
Dielectric strength	V_{AC}	200
Isolation resistance (03 schematic)	Ω	> 100M
Operating temperature range	$^{\circ}C$	-55 to +125

DIMENSIONS in inches (millimeters)



01 SCHEMATIC	GLOBAL MODEL	NUMBER OF RESISTORS	A (MAX.)	B	C (MAX.)
	CSC04	3	0.390 (9.91)	0.300 (7.62)	"A" profile = 0.195 (4.95) "B" profile = 0.295 (7.50)
	CSC05	4	0.490 (12.45)	0.400 (10.16)	
	CSC06	5	0.590 (14.99)	0.500 (12.70)	
	CSC07	6	0.690 (17.53)	0.600 (15.24)	
	CSC08	7	0.790 (20.07)	0.700 (17.78)	
	CSC09	8	0.890 (22.61)	0.800 (20.32)	
	CSC10	9	0.990 (25.15)	0.900 (22.86)	
	CSC11	10	1.09 (27.69)	1.00 (25.40)	
	CSC12	11	1.19 (30.23)	1.100 (27.94)	
03 SCHEMATIC	GLOBAL MODEL	NUMBER OF RESISTORS	A (MAX.)	B	C (MAX.)
	CSC04	2	0.390 (9.91)	0.300 (7.62)	"A" profile = 0.195 (4.95) "B" profile = 0.295 (7.50)
	CSC06	3	0.590 (14.99)	0.500 (12.70)	
	CSC08	4	0.790 (20.07)	0.700 (17.78)	
	CSC10	5	0.990 (25.15)	0.900 (22.86)	
	CSC12	6	1.19 (30.23)	1.100 (27.94)	
05 SCHEMATIC	GLOBAL MODEL	NUMBER OF RESISTORS	A (MAX.)	B	C (MAX.)
	CSC04	4	0.390 (9.91)	0.300 (7.62)	"A" profile = 0.195 (4.95) "B" profile = 0.295 (7.50)
	CSC05	6	0.490 (12.45)	0.400 (10.16)	
	CSC06	8	0.590 (14.99)	0.500 (12.70)	
	CSC07	10	0.690 (17.53)	0.600 (15.24)	
	CSC08	12	0.790 (20.07)	0.700 (17.78)	
	CSC09	14	0.890 (22.61)	0.800 (20.32)	
	CSC10	16	0.990 (25.15)	0.900 (22.86)	
	CSC11	18	1.09 (27.69)	1.00 (25.40)	
	CSC12	20	1.19 (30.23)	1.100 (27.94)	

MECHANICAL SPECIFICATIONS

Marking resistance to solvents	Permanency testing per MIL-STD-202, method 215
Solderability	Per MIL-STD-202, method 208E, RMA flux
Body	High alumina, epoxy coated
Terminals ⁽¹⁾	Solder plated leads

Note

⁽¹⁾ Coating meniscus meets class 2 requirements of IPC-A-610

STOCKED RESISTANCE VALUES IN Ω ("G" TOLERANCE)

Standard E-24 resistance values stocked; consult factory.

Many dual terminator resistance values stocked; consult factory.

IMPEDANCE CODES

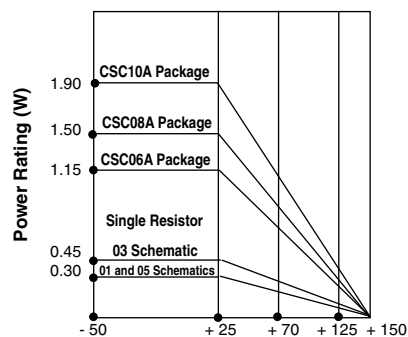
CODE	R ₁ (Ω)	R ₂ (Ω)	CODE	R ₁ (Ω)	R ₂ (Ω)
500B	82	130	141A	270	270
750B	120	200	181A	330	390
800C	130	210	191A	330	470
990A	160	260	221B	330	680
101C	180	240	281B	560	560
111C	180	270	381B	560	1.2K
121B	180	390	501C	620	2.7K
121C	220	270	102A	1.5K	3.3K
131A	220	330	202B	3K	6.2K

Note

- For additional impedance codes, refer to the Dual Terminator Impedance Code Table document (www.vishay.com/doc?31530)



"A" Profile



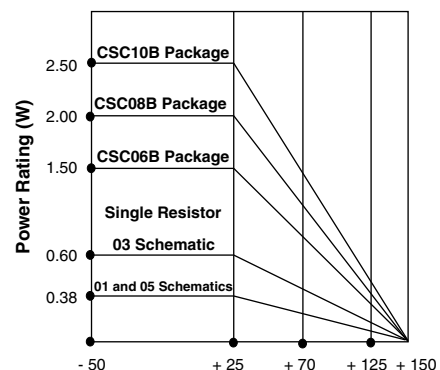
Derating

Ambient Temperature °C

"A" PROFILE +70 °C PACKAGE RATINGS

CSC12A	1.5 W
CSC11A	1.37 W
CSC10A	1.25 W
CSC09A	1.12 W
CSC08A	1.00 W
CSC07A	0.87 W
CSC06A	0.75 W
CSC05A	0.62 W
CSC04A	0.40 W

"B" Profile



Derating

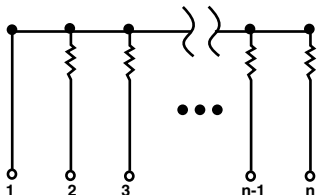
Ambient Temperature °C

"B" PROFILE +70 °C PACKAGE RATINGS

CSC12B	1.90 W
CSC11B	1.75 W
CSC10B	1.60 W
CSC09B	1.45 W
CSC08B	1.30 W
CSC07B	1.15 W
CSC06B	1.00 W
CSC05B	0.80 W
CSC04B	0.60 W

CIRCUIT APPLICATIONS

01 Schematic



Bussed

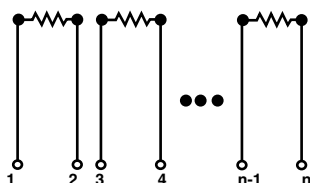
The CSCxxx01 single-in-line resistor networks provide the user with nominally equal resistors, each connected to a common pin (pin no. 1). Commonly used in the following applications:

- “Wired OR” pull-up
- Power gate pull-up
- MOS/ROM pull-up/pull-down
- Open collector pull-up
- TTL input pull-down
- TTL unused gate pull-up

Note

- “A” profile standard, “B” profile available

03 Schematic



Isolated

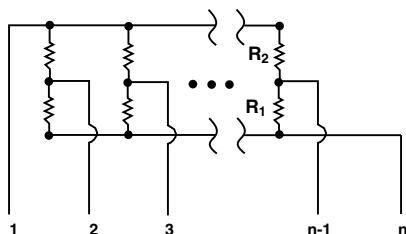
The CSCxxx03 single-in-line resistor networks provide the user with nominally equal resistors. Each resistor is isolated from all others. Commonly used in the following applications:

- “Wired OR” pull-up
- Power driven pull-up
- Power gate pull-up
- Line termination
- Long-line impedance balancing
- LED current limiting
- ECL output pull-down
- TTL input pull-down

Note

- “A” profile standard, “B” profile available

05 Schematic



Dual Terminator

The CSCxxx05 circuits contain series pairs of resistors. Each series pair is connected between two common lines. The junction of these resistor pairs is connected to the input terminals. The 05 circuits are designed for TTL dual-line termination and pulse squaring.

Note

- “A” profile standard, “B” profile available

PERFORMANCE

TEST	CONDITIONS	MAX. ΔR (TYPICAL TEST LOTS)
Thermal shock	5 cycles between -65 °C and +125 °C	$\pm 0.50 \% \Delta R$
Short time overload	2.5 x rated working voltage, 5 s	$\pm 0.25 \% \Delta R$
Low temperature operation	45 min at full rated working voltage at -65 °C	$\pm 0.25 \% \Delta R$
Moisture resistance	240 h with humidity ranging from 80 % RH to 98 % RH	$\pm 1.00 \% \Delta R$
Resistance to soldering heat	Leads immersed in +350 °C solder to within 1/16" of body for 3 s	$\pm 0.25 \% \Delta R$
Shock	Total of 18 shocks at 100 g's	$\pm 0.25 \% \Delta R$
Vibration	12 h at maximum of 20 g's between 10 Hz and 2000 Hz	$\pm 0.25 \% \Delta R$
Load life	1000 h at +70 °C, rated power applied 1.5 h “ON”, 0.5 h “OFF” for full 1000 h period; derated according to the curve	$\pm 1.00 \% \Delta R$
Terminal strength	4.5 pound pull for 30 s	$\pm 0.25 \% \Delta R$
Insulation resistance	10 000 M Ω (minimum)	-
Dielectric withstanding voltage	No evidence of arcing or damage (200 V _{RMS} for 1 min)	-



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